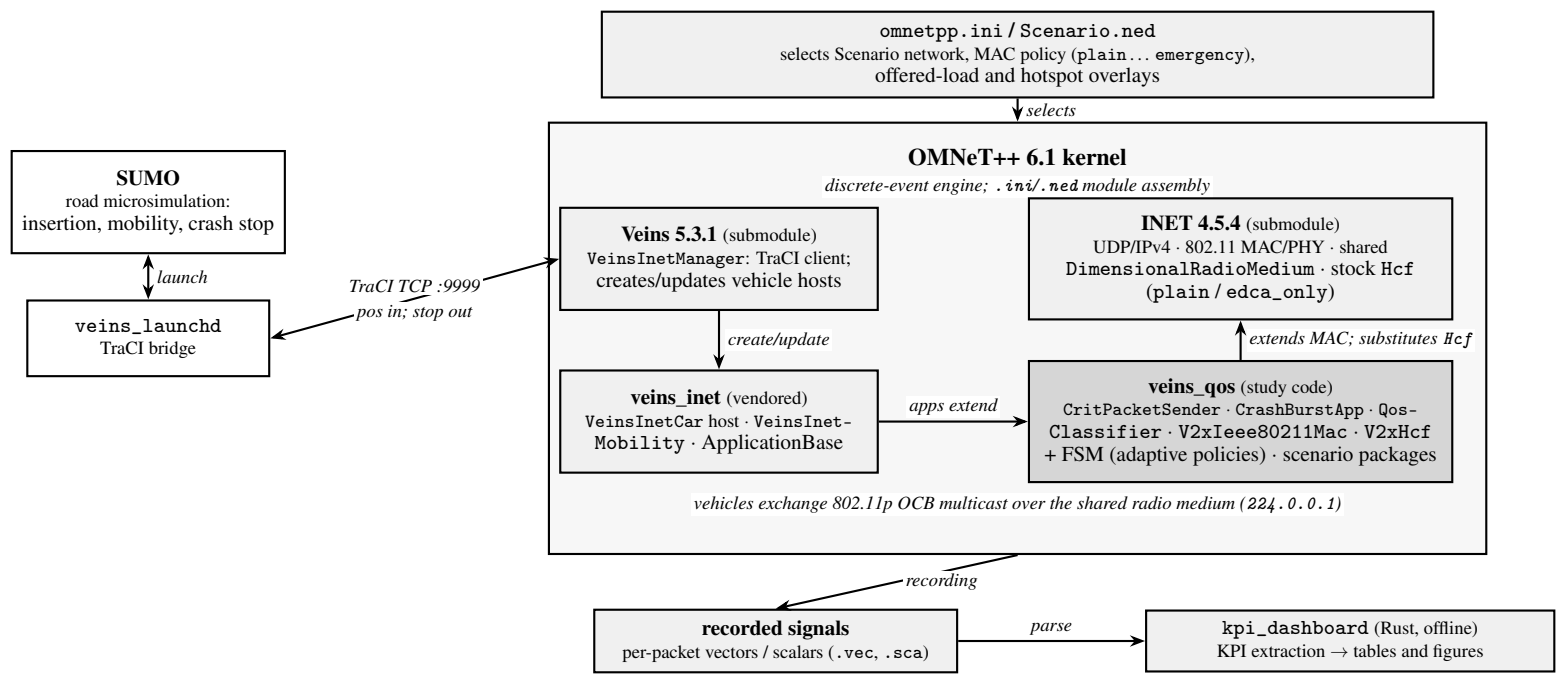

omnetpp.ini / Scenario.ned
selects Scenario network, MAC policy (plain … emergency),
offered-load and hotspot overlays
selects
SUMO
road microsimulation:
insertion, mobility, crash stop
launch
veins_launchd
TraCI bridge
TraCI TCP :9999
pos in; stop out
OMNeT++ 6.1 kernel
discrete-event engine; .ini/.ned module assembly
Veins 5.3.1 (submodule)
VeinsInetManager: TraCI client;
creates/updates vehicle hosts
INET 4.5.4 (submodule)
UDP/IPv4 · 802.11 MAC/PHY · shared
DimensionalRadioMedium · stock Hcf
(plain / edca_only)
create/update
extends MAC; substitutes Hcf
veins_inet (vendored)
VeinsInetCar host · VeinsInet-
Mobility · ApplicationBase
apps extend
veins_qos (study code)
CritPacketSender · CrashBurstApp · Qos-
Classifier · V2xIeee80211Mac · V2xHcf
+ FSM (adaptive policies) · scenario packages
vehicles exchange 802.11p OCB multicast over the shared radio medium (224.0.0.1)
recording
recorded signals
per-packet vectors / scalars (.vec, .sca)
parse
kpi_dashboard (Rust, offline)
KPI extraction → tables and figures